

Evaluating a Deterministic Validator Plus Single-Iteration Repair Pipeline for LLM-Generated Network Configurations

Autonomous AI Researcher
CNSM 2026 Agentic AI Researcher Track

Abstract—We preregistered and executed a paired confirmatory experiment (N=300 paired intent→configuration tasks; pre-registration id cand-2026-001-A-prereg-v1, preregistration SHA-256 archived) to evaluate whether a minimal guarded pipeline—single-shot LLM generation (declared model: gpt-5-mini) followed by a frozen deterministic configuration validator and at most one automated repair iteration—reduces first-pass misconfiguration relative to plain single-shot generation. Execution completed 600 episodes and used 301 model calls. Observed outcomes: baseline = 299/300 valid (99.667%); guarded = 300/300 valid (100.0%); paired counts n11=299, n10=1, n01=0, n00=0. Exact McNemar two-sided $p = 1.00$; absolute paired difference = 0.0033333; paired bootstrap 95% CI = [0.00, 0.01] (10,000 resamples, seed 1729). The single permitted automated repair corrected the sole invalid baseline output (repair success 1/1; Clopper–Pearson 95% CI = [0.025, 1.0]). The preregistered planning assumption used for power calculations (planned baseline pass rate = 0.40) was contradicted by the observed near-ceiling baseline, removing statistical headroom for the preregistered $\geq 50\%$ relative reduction target. We report preregistered design choices, execution accounting, confirmatory statistics, a post-hoc inferential sensitivity bound tied to the observed contingency counts, and artifact-grounded recommendations for future NetOps confirmatory evaluations. All run artifacts and analysis outputs are archived in the execution manifest referenced in the Disclosure Statement.

I. INTRODUCTION

Translating operator intent to device configuration is a core NetOps task. Large language models (LLMs) have been proposed as intent→configuration translators that can accelerate operations, but probabilistic generation risks misordering, omission, and unsafe commands. Recent literature advocates hybrid patterns that pair probabilistic generators with deterministic validators/simulators and bounded repair loops as operational floor-safety nets. However, rigorous preregistered evaluations that quantify how tightly bounded guarded pipelines change first-pass misconfiguration rates under controlled sampling semantics are limited.

We implement and preregister a narrowly scoped paired experiment to evaluate the operational effect of a minimal guarded pipeline composed of: (1) a single shared LLM candidate per task (single-shot generation), (2) a frozen deterministic validator performing canonicalization and rule/dependency checks, and (3) at most one automated repair attempt when the validator reports violations. The core research question is: Can this minimal guarded pipeline reduce first-pass misconfiguration rate relative to plain single-shot generation under a

controlled paired design? Our contributions are (i) a preregistered paired experiment with deterministic seed generation and archived per-task artifacts, (ii) confirmatory statistics derived from those artifacts, (iii) a compact post-hoc sensitivity quantification tied to the observed contingency counts, and (iv) artifact-grounded recommendations for future NetOps confirmatory evaluations.

II. RELATED WORK

Recent work on LLMs for NetOps and intent-driven configuration has produced numerous system demonstrations and surveys documenting both promise and concrete failure modes. Empirical demonstrations of intent→configuration translation report that LLMs can synthesize plausible device configuration templates in lab and testbed settings, yet they frequently surface operationally relevant errors such as command misordering, omission of required safety steps, and overconfident assertions that lack mechanical verification (example system evaluation and discussion: Nguyen et al. 2024 [10.1002/nem.2313]). Complementary survey and review articles consolidate these observations and emphasize the need for grounded verification and governance when applying LLMs to critical network tasks [NIST guidance and surveys, 10.6028/nist.ai.100-2e2023].

Parallel technical threads motivate guarded architectures. Work in adjacent domains (clinical QA, code repair, and general RAG/verification research) shows measurable reductions in factual errors when probabilistic generators are paired with retrieval, deterministic checks, or repair loops; these findings have been ported by several NetOps proposals arguing for generate→verify→repair pipelines rather than blind single-shot deployment [e.g., comparative/procedural studies and framework discussions in the literature pool, 10.36227/techrxiv.176784505.56675782/v1; 10.2139/ssrn.6608518]. Architectural proposals for intent-aware automated remediation and multi-agent NetOps orchestration similarly argue that deterministic validators and bounded repair/backout policies are central to safe agentic NetOps workflows (see recent agentic NetOps proposals and intent-translation demonstrations, e.g., 10.36227/techrxiv.177004277.71795055/v1).

Deterministic network verification provides concrete algorithmic foundations for validators used in guarded pipelines. Prior work on specification-based and header-space analyses demonstrates tractable static checks for reachability, ac-

cess control, and update-safety properties without requiring live device execution; such algorithms can be integrated to canonicalize generated artifacts into normalized ASTs and to flag violations that trigger repair logic [Beckett et al., 2017 foundation for incremental/specification checks, 10.1145/3098822.3098834]. These deterministic checks differ fundamentally from empirical or emulation tests: they provide provable, rule-based pass/fail decisions for a defined specification, but they do not by themselves measure runtime control-plane or packet-level effects unless combined with higher-fidelity simulation or emulation.

Limitations in the prior literature motivate the present work. Many demonstrations and proposals are system-level or exploratory rather than preregistered confirmatory experiments; reported improvements from RAG, verifier-assisted generation, or agentic architectures often lack paired within-task statistical controls that isolate the incremental effect of a validator+repair floor on the same generated candidate. Moreover, published comparisons rarely quantify how sampling semantics (shared candidate versus independent draws) and repair budgets affect the number of informative discordant pairs available to paired binary tests. Surveys and architectural reviews therefore call for reproducible evaluation harnesses that combine deterministic validators with controlled sampling semantics and explicit inferential plans [10.6028/nist.ai.100-2e2023].

Our study addresses a tightly scoped portion of this gap by executing a preregistered paired experiment that (a) fixes `shared_initial_candidate` semantics to measure remediation capacity on a single generated plan, (b) uses a frozen deterministic validator to define a binary success criterion tied to intent constraints, and (c) limits repair budget to a single automated iteration. This design directly tests the operational claim that a minimal validator+repair floor reduces first-pass misconfiguration, and it surfaces the inferential consequences of high baseline pass rates and shared-candidate pairing for confirmatory McNemar-style testing.

III. METHODOLOGY

Preregistration and confirmatory hypotheses: The experiment was preregistered under id `cand-2026-001-A-prereg-v1` before any final outcomes were observed. Two confirmatory hypotheses were registered: H1 — the guarded pipeline reduces first-pass misconfiguration rate by $\geq 50\%$ (relative) versus single-shot generation; H2 — one or fewer automated repair iterations recover $\geq 75\%$ of initially invalid configurations. The preregistration explicitly documented the numeric planning assumption used for power calculations: planned baseline pass rate = 0.40. This planning assumption informed the targeted sample size and the preregistered effect magnitude; because it is central to the inferential design we report it here in the Methods for transparent interpretation.

Design and sampling semantics: We used a paired within-task design with a fixed deterministic seed list of 300 tasks produced by a frozen task generator. Each seed produced an intent and an associated canonical reference

configuration; seeds were included deterministically unless a preregistered parser exclusion rule applied (such replacements are recorded in the archived manifest). For each seed the same single LLM candidate was used in both arms (`shared_initial_candidate` semantics): the generator produced one configuration (shared candidate), which served as the baseline response and was the starting point for the guarded arm’s validation and possible repair. Planned episodes = 600 (two episodes per seed), `initial_generation_calls_per_task` = 1, `maximum_repair_calls_per_task` = 1. The shared-candidate pairing was preregistered to isolate the incremental effect of deterministic validation and bounded repair on a single proposed plan; as noted below, this choice increases within-pair correlation and reduces the number of statistically informative discordant pairs available to paired tests compared with an independent-draws design.

Model, resource constraints, and provider-call policy: The declared generation model in the execution contract was `gpt-5-mini`. The execution contract limited total model calls to a planned maximum of 600 (one initial generation per seed across 300 seeds plus up to one repair per seed). Provider call failures are handled per the preregistered retry policy (single retry allowed for provider/infrastructure failures); the primary missingness treatment drops pairs only if both calls in a pair failed (`'complete_pair_primary'`). These operational constraints are part of the preregistration because they affect the realized information and the primary analysis set.

Validator canonicalization, rules, and scoring semantics: The frozen deterministic validator implemented canonicalization to a normalized configuration AST and applied a suite of static checks derived from explicit workflow and dependency policies encoded in the task payloads. The scoring rule used across the experiment was full intent-constraint validation: a generated configuration is scored as success only when (a) the validator’s `normalized_configuration` satisfies the task’s `required_sequence` and `expected_final_state` constraints, and (b) all preregistered workflow policies (e.g., `'must_be_down_for_fields'`, `'final_group_constraints'`, and `'minimum_active_in_groups'`) are satisfied. The validator’s trace records `parsed_command_count`, `required_command_count`, `observed_touched_field_count`, `violation_codes`, and a boolean valid flag; these elements were the operational primitives used to trigger repair and to compute per-task scores. Canonicalization enables deterministic AST equality and normalized-string comparisons used in contamination detection and exact matching checks.

Repair adapter behavior and steering: The preregistered automated repair adapter is invoked only when the validator reports violations on the shared candidate. The repair adapter receives the shared candidate plus structured violation codes and is restricted to a single automated repair attempt per episode. Its objective is minimal, targeted editing of the shared candidate sufficient to remove the reported violations while preserving as much of the original plan as permitted by task constraints. Repair prompts and repair responses were

archived per episode; the repair outcome was revalidated by the same deterministic validator to produce the final guarded decision. Because the repair budget was intentionally bounded to one call, repair-effectiveness estimands are conditional on the initially invalid subset and must be interpreted with that budget constraint in mind.

Contamination checks, exclusions, and replacement rules: The preregistration included deterministic contamination detection based on exact normalized-AST equality and deterministic near-duplicate lexical thresholds; any exact duplicates between task targets and transformation/unit-test artifacts would trigger deterministic exclusion and deterministic replacement from a preregistered reserve set. The primary analysis uses the decontaminated paired set produced by these rules; sensitivity analyses are specified to report results on the full set when flags arise. All exclusions, replacements, and contamination flags are recorded in the archived manifest per the preregistered plan.

Primary estimand and statistical analysis plan: The primary estimand was the `paired_success_rate_difference_guarded_minus_baseline` (absolute difference in per-task success probability under guarded vs baseline). The preregistered primary test was the exact McNemar two-sided test at $\alpha=0.05$ on paired binary outcomes; paired bootstrap percentile 95% CIs for the absolute paired difference were preregistered (10,000 resamples, explicit seed). Secondary analyses included exact binomial CIs for repair success conditional on initial failure, stratified paired checks by task difficulty and constrained vendor templates, and sensitivity treatments for failed provider calls (treat as failures or exclude per the preregistered missingness plan). The preregistration contained a power/precision plan that translated the planned baseline pass rate = 0.40 and a target relative reduction ($\geq 50\%$ relative reduction in failure rate) into a required N under conservative discordant assumptions; that planning arithmetic is reported in the preregistration record and is discussed in the manuscript’s interpretation because the observed baseline diverged from that assumption.

Sampling-semantics tradeoffs and inferential consequences: We preregistered the `shared_initial_candidate` semantics to align the experiment with operational workflows that commonly evaluate and remediate a single proposed plan. This design reduces cross-arm generation variance (the same generated plan is evaluated and potentially repaired) but increases within-pair correlation: paired discordance counts are limited to cases where the validator+repair pipeline changes the acceptability of that single candidate. Consequently, the number of informative discordant pairs available to McNemar inference is typically smaller than under independent-draws designs and must be accounted for when translating planned effect sizes into target sample sizes. The preregistration explicitly encoded this tradeoff in its power calculations; we reiterate it here to make the methodological rationale and its inferential implications explicit in the Methods rather than only in the Discussion.

Reproducibility and archival primitives: The experiment preregistered and archived all deterministically generated seeds, the task generator configuration, the scoring rules, validator trace formats, repair prompt templates, the analysis code, and the execution manifest. Key reproducibility primitives include the normalized AST canonicalization used by the validator (the basis for deterministic equality checks), the pair sampling semantics (shared candidate per pair), the maximum repair budget (one per episode), the missingness/retry policy, and the primary exact McNemar test with bootstrap CI parameters (10,000 resamples, explicit seed). These artifacts are enumerated in the execution manifest and analysis bundle referenced in the Disclosure Statement and constitute the authoritative sources for the numeric counts and inferential inputs reported in this paper.

IV. RESULTS

Execution accounting and artifact provenance (archived): The preregistered run completed all planned episodes: planned episodes = 600, completed episodes = 600, complete analyzed pairs = 300. Model calls consumed = 301 total (300 shared initial generation calls + 1 repair invocation). The archived deterministic reconciliation/provider audit records show no provider or infrastructure failures and no deterministic exclusions for contamination under the preregistered checks. These execution counts and stage summaries are recorded in the archived execution manifest and analysis bundle and are the direct source for all numeric values reported below.

Primary per-condition outcomes (archived counts): Baseline (single-shot) success count = 299/300 (99.6667%); Guarded (validator + ≤ 1 repair) success count = 300/300 (100.0%). The paired contingency table derived from the archived analysis artifact is: n_{11} (both success) = 299; n_{10} (guarded success, baseline fail) = 1; n_{01} (guarded fail, baseline success) = 0; n_{00} (both fail) = 0. Missingness summary in the archive reports zero failed or unscorable episodes in either arm under the preregistered missingness treatment.

Confirmatory inferential results (preregistered analyses executed on archived artifacts): The exact McNemar two-sided test on the observed contingency yields $p = 1.00$. The observed absolute paired difference (guarded - baseline success rate) = $1/300 \approx 0.0033333$. Paired bootstrap percentile 95% CI for the absolute paired difference = [0.00, 0.01] (10,000 resamples, bootstrap seed = 1729), as computed in the archived analysis artifact. Repair outcomes: the single preregistered automated repair invocation succeeded on the sole initially invalid baseline output (repair success 1/1). The exact Clopper-Pearson 95% CI for a 1/1 repair success is [0.025, 1.0], reflecting the large uncertainty from a single observed repair case.

Why the McNemar $p = 1.00$ and what the discordant count implies: Exact McNemar inference is driven entirely by the discordant pair counts (n_{10} and n_{01}). With only one discordant pair observed ($n_{10}+n_{01} = 1$), the two-sided exact test cannot reject the null of no difference; the minimal nonzero discordant total yields an exact two-sided p of 1.00 under the test implementation used in the preregistered analysis. Equivalently, the

discordant total places a hard upper bound on the resolvable absolute paired difference equal to discordants / 300 = $1/300 \approx 0.00333$. Put practically, the observed near-ceiling baseline left essentially no statistical headroom to detect the preregistered $\geq 50\%$ relative reduction (which, given the preregistration’s planning baseline = 0.40, mapped to an absolute change on the order of +0.30). Detecting an absolute +0.30 with $N=300$ would require on the order of tens of discordant pairs (the preregistration noted ~ 90 net discordant pairs aligned with the planned effect size under conservative assumptions); the archived discordant count of one is far below that requirement. These limits and numeric bounds are computed from the archived contingency artifact and bootstrap outputs.

Representative validator and diagnostic traces (artifact-grounded): For each episode the archived validator trace contains structured diagnostics used for scoring: `normalized_configuration`, `parsed_command_count`, `required_command_count`, `observed_touched_field_count`, `violation_codes`, `validation_before/after` states, and a `repair_applied` flag. In the solitary baseline failure the validator trace documents the missing required command sequence and the canonicalized configuration before and after the automatic repair; the post-repair canonicalized configuration satisfied the full intent constraints per the validator. These per-task traces (validator decision fields and repair flags) are the primary provenance for the contingency counts and are archived alongside the analysis artifacts.

Sensitivity and interpretive bounds (post-hoc, artifact-grounded): Because the observed discordant total is one, the maximal absolute improvement observable in this dataset equals $1/300$. The paired bootstrap CI [0.00, 0.01] and the exact McNemar p together reflect that any larger effect size is incompatible with the observed discordant evidence. Operationally, this implies that confirmatory designs aiming to test large relative improvements must either (a) calibrate the task bank to produce a higher baseline failure prevalence or (b) adopt independent-draws sampling (increasing discordants at the cost of within-pair control) — choices that should be incorporated into power calculations. These precise numerical bounds are reproduced in the archived `paired_contingency_table.csv` and `paired_difference_figure.svg` contained in the analysis bundle.

Reproducibility and archived artifacts: All numerical counts, the bootstrap procedure (seed = 1729, resamples = 10,000), the exact-test implementation, and every per-task validator trace and repair prompt/response are included in the archived execution manifest and analysis bundle. The archived analysis artifacts (condition summary, paired contingency table, bootstrap outputs, and deterministic reconciliation/provider audit) reproduce the figures and numbers above when loaded with the provided analysis executor. Readers who wish to reproduce the exact inferential outputs can consult the execution manifest and the bundled analysis artifacts referenced in the Disclosure Statement.

Concise summary of the results section: The guarded pipeline corrected the single observed invalid baseline output

(artifact-grounded repair success = 1/1) but, because the baseline pass rate was far higher than the preregistration planning assumption, the run produced only one discordant pair and therefore lacked statistical information to verify the preregistered $\geq 50\%$ relative reduction. The run nevertheless demonstrates end-to-end execution, archival, and artifact-grounded diagnostics for a generate→validate→repair NetOps pipeline and clarifies concrete sample-size and sampling-semantic constraints necessary for informative confirmatory inference in future studies.

V. DISCUSSION

We expand the Discussion to emphasize concrete, artifact-grounded interpretation, diagnostic lessons from the run, and practical guidance for designing future confirmatory NetOps studies.

Quantitative interpretation tethered to archived counts. The archived contingency ($n_{11}=299$, $n_{10}=1$, $n_{01}=0$, $n_{00}=0$) directly constrains inferential outcomes: the paired difference equals $(n_{10} - n_{01})/N = 1/300 \approx 0.00333$, and any two-sided exact McNemar test necessarily conditions on the observed discordant count (here 1). With only one discordant pair, the sampling distribution under the null contains insufficient extreme outcomes to produce a small p -value; operationally this is why the exact McNemar test returns $p = 1.00$ for our observed pattern. The paired bootstrap CI [0.00, 0.01] (10,000 resamples, seed 1729) expresses the same limitation in interval form: the data are consistent only with extremely small absolute improvements given the observed discordance. These numerical facts are archived and reproduce directly from the execution manifest and analysis artifacts.

Why preregistered planning assumptions matter. Our preregistered power plan assumed a baseline pass rate = 0.40; that planning choice drove an expectation of many discordant pairs under a substantial relative reduction target. The actual baseline (0.9967) collapses that expectation. The practical lesson is that power planning for generate-verify repair experiments must explicitly consider plausible baseline prevalences and the chosen pairing semantics (shared candidate vs independent draws), because both jointly determine expected discordant counts. A simple arithmetic relation is useful when translating target absolute paired differences into discordant-pair targets: absolute paired difference = $(n_{10} - n_{01})/N$, so for $N=300$ an absolute increase of 0.30 requires $(n_{10} - n_{01}) \approx 90$ net discordant pairs favoring guarded outputs. The archived artifacts provide the observed (near-zero) discordant count and therefore an artifact-grounded explanation of why the preregistered $\geq 50\%$ relative reduction could not be detected.

Pairing semantics: operational rationale and inferential tradeoffs. We preregistered a `shared_initial_candidate` design to emulate a realistic operational workflow: operators often review and repair a single candidate plan rather than comparing independently drawn proposals. This choice improves ecological validity for the intended operational claim (effect of validator+repair on a single generated plan) but reduces statistical signal because within-pair correlation lowers expected

discordant counts. In contrast, an independent-draws design yields larger expected discordance for a given marginal success probability and thus greater power for McNemar-style paired tests; however, it addresses a different operational question (expected net benefit of switching from one generator draw to another) and increases heterogeneity. Future preregistrations should explicitly link the pairing semantics to the estimand of operational interest and incorporate that link into power and sample-size computations.

Repair effectiveness as a conditional estimand and sample-size implications. H2 framed repair success conditional on initial failure. The run observed repair success 1/1, but that conditional estimate is extremely imprecise (Clopper–Pearson 95% CI = [0.025, 1.0]). Practically, when repair effectiveness is a primary operational claim, study designers should power the experiment for the conditional probability $P(\text{repair succeeds} \mid \text{initial failure})$ by ensuring a planned number of initial failures (not merely overall N). This can be achieved either by calibrating task difficulty to increase baseline failure prevalence or by targeted adversarial augmentation of tasks so that the expected number of initial failures meets the sample-size requirements for the desired CI width.

Diagnostics enabled by deterministic validator traces. The frozen deterministic validator produced structured per-task traces (normalized_configuration, parsed_command_count, required_command_count, violation_codes, repair_applied). These traces are essential for post-hoc failure-mode analysis: they identify whether failures arise from syntax/canonicalization mismatches, missing commands, dependency-order violations, or preserved-state violations (e.g., touching protected interfaces). In our run the single baseline failure was diagnosed as a required-sequence omission by the validator and repaired by applying the preregistered repair adapter; the repair trace shows the minimal edit needed to satisfy intent constraints. Maintaining such per-task structured traces is a practical requirement for explainable NetOps guardrails and for reproducible post-mortems.

Construct validity and the role of dynamic emulation. Our validator performs deterministic static checks and canonicalization rather than executing configurations in devices or emulators; this design increases safety by avoiding live changes but limits construct validity with respect to runtime semantic properties (control-plane convergence, routing dynamics, packet-level reachability under transient state). If operational claims require guarantees about runtime consequences, future studies should integrate emulation or simulation stages that exercise control-plane and dataplane effects and record dynamic assertions (e.g., convergence time, transient reachability). The archived manifest documents that our results are strictly conditional on the validator’s static semantics.

Design recommendations grounded in observed artifacts. Based on the run artifacts and contingency counts, we recommend three concrete design levers for future confirmatory NetOps evaluations: 1) Calibrate task difficulty or perform adversarial augmentation to target a nontrivial baseline failure

prevalence or a predetermined discordant-pair count aligned with the planned absolute/relative effect size. Use deterministic task generators with adjustable difficulty knobs and preregister the target discordant count. 2) Explicitly preregister pairing semantics and repair budgets and incorporate their expected effects on discordant counts into power calculations; if the operational question aims to evaluate remediation of a single plan, prefer shared-candidate pairing but then set task difficulty to produce more initial failures. 3) When runtime semantics matter, add an emulation/simulation validation stage and preregister how emulator outcomes map to success/failure to avoid post-hoc redefinition of the estimand. All of these recommendations are traceable to artifacts in the execution manifest (task difficulty fields, validator violation codes, and the paired contingency table) and are therefore actionable.

Threats to validity revisited with artifact examples. Internal validity: the shared_candidate design and deterministic replacement rules are preregistered but reduce the number of informative discordant pairs; this is evident in the archived stage_counts and provider_call audit. Construct validity: static validator checks may miss dynamic failure modes; the validator traces explicitly show which checks were applied and which dynamic behaviors were not exercised. External validity: the synthetic task bank and constrained vendor templates are enumerated in the task manifest and limit generalization across vendor heterogeneity and production topologies. Conclusion validity: the paucity of discordant pairs (one) creates high sampling uncertainty for large preregistered effects; the paired bootstrap and exact McNemar results in the archived analysis artifacts document this limitation quantitatively.

Operational implications for deploying guarded pipelines. The run shows that a frozen deterministic validator plus a bounded repair loop can correct at least some invalid candidates without human intervention (the single observed correction). However, whether such a pipeline materially reduces first-pass misconfiguration in production depends on the baseline error prevalence and the validator’s coverage of real failure modes. When baseline error is already rare, the primary operational value of a validator+repair may be as a safety net that prevents the rare catastrophic mistake rather than as a mechanism to substantially increase throughput. Organizations should therefore align evaluation goals—error-reduction vs catastrophic-risk mitigation—with study design choices (task bank selection, pairing semantics, emulation inclusion) and with the operational acceptance thresholds that govern whether automated repair may be applied without human approval.

Reproducibility and archival practice. This run demonstrates the utility of a full artifact archive: per-task inputs, normalized responses, validator traces, repair prompts/responses, the paired contingency table, and the analysis code together enable independent recomputation of reported counts and intervals. The execution manifest documents model_calls_used = 301 and repair_stage invocations = 1; these accounting numbers are minimal but sufficient to audit the reported outcomes. Future confirmatory NetOps work should adopt similar archival granularity to permit external verification and subsequent

meta-analysis.

VI. LIMITATIONS

- Synthetic task bank and constrained vendor-template scope limit external validity to broad production environments.
- Single LLM (gpt-5-mini) and a single validator implementation restrict generality across model families and validator designs.
- The preregistered power plan assumed a baseline pass rate = 0.40; the observed baseline (299/300) contradicts that assumption and removed headroom to detect the preregistered $\geq 50\%$ relative reduction.
- Sampling semantics (shared_initial_candidate) reused the same initial candidate across paired arms, increasing within-pair correlation and reducing discordant pairs available to McNemar inference.
- Validation used deterministic configuration/constraint checks rather than full dynamic emulation of runtime semantic effects; actual runtime consequences of configurations were not executed and remain unmeasured.
- H2 repair-success evidence is based on a single initially invalid case (1/1 $\rightarrow$ 100%), yielding a wide Clopper-Pearson 95% CI = [0.025, 1.0]; larger counts of initial failures are required to estimate repair effectiveness precisely.

VII. CONCLUSION

We executed a preregistered paired experiment (N=300 pairs) comparing single-shot LLM generation (gpt-5-mini) to a guarded pipeline consisting of a frozen deterministic validator and a single automated repair iteration. Baseline single-shot generation yielded 299/300 valid outputs and the guarded pipeline yielded 300/300 valid outputs. The single automated repair corrected the lone invalid baseline output, but the observed near-ceiling baseline contradicted the preregistered planning assumption (baseline pass rate = 0.40) and removed statistical headroom to detect the preregistered $\geq 50\%$ relative reduction. Future confirmatory NetOps evaluations should (a) calibrate task difficulty to produce sufficient initial failures or target discordant counts, (b) preregister pairing semantics and repair budgets explicitly, and (c) include emulation to measure runtime semantic effects when operational deployment claims are intended. All raw outputs, validator traces, repair traces, the execution manifest, and analysis artifacts are archived and enumerated in the Disclosure Statement to permit reproduction of the reported counts and intervals.

DISCLOSURE STATEMENT

This study was executed autonomously after the autonomy lock under the archived master prompt (provenance/master_prompt.txt; SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba). Human role: the Research Director selected the topic family (Generative AI and LLMs for NetOps) and approved the master prompt prior to the autonomy lock; no manual

human editing of manuscript technical content, figures, tables, or references occurred after the lock. Preregistration: cand-2026-001-A-prereg-v1 (the preregistration record was created before observing final outcomes; preregistration SHA-256 is recorded in the execution manifest). Declared model and tooling: gpt-5-mini (declared_model_version), adapter family hosted_netops_gvr_v1, frozen deterministic validator/repair adapter deterministic_netops_validator_v5. Execution accounting (archived): planned episodes = 600, completed episodes = 600, complete pairs = 300, model_calls_used = 301 (300 shared initial generation calls + 1 repair invocation), repair-stage invocations = 1. The execution manifest and analysis artifact bundle enumerate all raw model responses, per-task validator traces (normalized configurations, parsed_command_count, required_command_count, violation_codes, repair_applied flags), repair prompts/responses, execution logs, and analysis artifacts. The archived execution manifest and analysis bundle are the authoritative source for the numeric counts reported in this manuscript. The autonomous pipeline performed literature search, preregistration, execution, analysis, AI-peer-review style checks, and manuscript drafting autonomously after the autonomy lock in accordance with the CNSM Agentic AI Researcher track requirements. The preregistered planning assumption used in power calculations (planned baseline pass rate = 0.40) is explicitly reported here because it materially affects interpretation of the confirmatory result.

REFERENCES

- [1] Nguyen Tu, Sukhyun Nam, James Won-Ki Hong, "Intent-Based Network Configuration Using Large Language Models", 2024, 10.1002/nem.2313
- [2] Oghenekeno Hilkhiah Okula, "The Era of AI Agents for Network Engineering": Intent-Aware Auto Remediation for Network Disruption or Failures Using AI Agents, Large Language Models (LLM) and Model Context Protocol (MCP) Mechanisms", 2026, 10.36227/techrxiv.177004277.71795055/v1
- [3] Kunal Dhanda, "MedRAG: Retrieval-Augmented Generation for Medical QA-Comparing Base and RAG-Augmented LLM Performance on Evidence from Peer-Reviewed Clinical Research", 2026, 10.2139/ssrn.6608518
- [4] Goutam Adwant, Manjari Srivastav, "Evidence Coverage Evaluator: A Comprehensive Framework for Assessing Grounding Quality in Retrieval-Augmented Generation Systems", 2026, 10.36227/techrxiv.176784505.56675782/v1
- [5] Ryan Beckett, Aarti Gupta, Ratul Mahajan, David Walker, "A General Approach to Network Configuration Verification", 2017, 10.1145/3098822.3098834
- [6] Apostol Vassilev, Alina Oprea, Alie Jean Fordyce, Hyrum S. Anderson, "Adversarial machine learning :", 2024, 10.6028/nist.ai.100-2e2023